

## **TREE**

Definition: A **Tree** is a connected acyclic undirected graph. It is a discrete structure that represents hierarchical relationships between individual elements or nodes. A tree in which a parent has no more than two children is called a binary tree.

### **Properties of a Tree**

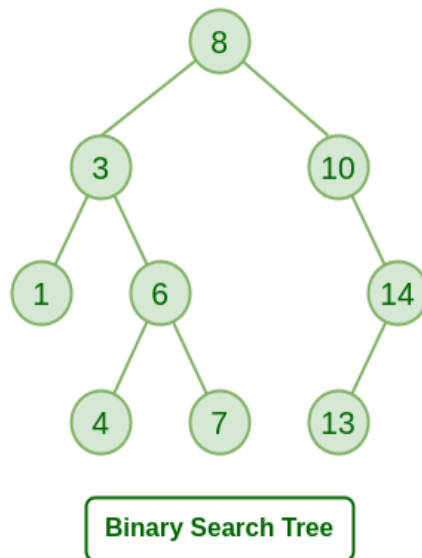
In discrete mathematics, when discussing trees (specifically rooted trees), several key terminologies are used to describe the relationships between nodes and edges within the tree structure. These include:

- **Root:** The topmost node in a tree, from which all other nodes descend.
- **Node:** A fundamental unit in a tree, representing a point or location.
- **Edge:** A line connecting two nodes, representing a relationship between them.
- **Parent:** A node that is directly above another node in the tree structure.
- **Child:** A node that is directly below another node in the tree structure.
- **Sibling:** Nodes that share the same parent.
- **Ancestor:** A node that is an ancestor of another node if it is on the path from that node to the root.
- **Descendant:** A node that is a descendant of another node if the other node is an ancestor of it.
- **Leaf:** A node with no children.
- **Internal Node:** A node that is not a leaf.
- **Binary Tree:** A tree where each internal node has at most two children.

## **BINARY SEARCH TREE**

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• A **Binary Search Tree (or BST)** is a data structure used in computer science for organizing and storing data in a sorted manner. Each node in a Binary Search Tree has at most two children, a left child and a right child, with the left child containing values less than the parent node and the right child containing values greater than the parent node. This hierarchical structure allows for efficient searching, insertion, and deletion operations on the data stored in the tree.



Problem:

1. Given: The following sequence of numbers:  
Sequence : **{50, 30, 70, 20, 40, 60, 80}**

Task: Insert the items into an empty binary search tree in order.

Algorithm

- Insert 50 → root
  - Insert 30 → left of 50
  - Insert 70 → right of 50
  - Insert 20 → left of 30
  - Insert 40 → right of 30
  - Insert 60 → left of 70
  - Insert 80 → right of 70
2. Given: The following sequence of numbers:

Sequence: {25, 15, 50, 10, 22, 35, 70}

Task: Write the algorithm and insert the items into an empty binary search tree in order.

3. Given: The following sequence of numbers:

Sequence: {30, 40, 23, 58, 48, 26, 11, 13}

Task: Write the algorithm and insert the items into an empty binary search tree in order.